

Kevin Hertlein

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PERSONAL STATEMENT

Software Engineer with 8+ years of experience working on mixed signal/RF embedded platforms focused on board support and driver development, with a strong preference for hands-on lab environment

EDUCATION

University of Florida, Gainesville, FL, **Graduation Date** May 2017

Bachelor of Science in Computer Engineering **GPA** 3.88

Programming and Scripting Languages C, Python, C++, LabVIEW, C#, JavaScript, VBA

WORK EXPERIENCE

General Motors, Austin, Texas - *Digital Vehicle Experience - Embedded Radio and Connectivity*

Senior Software Engineer, Aug 2025 - Present

- Led integration of production MCU firmware (RTOS, AUTOSAR, C) into board support package, enabling firmware build, packaging, and board flashing as part of the Yocto distribution
- Contributed to bring-up of 5G/Wi-Fi ECUs through peripheral (CAN/I2C/SPI) validation, SWD/JTAG enablement, and development of automated board flashing tools (Python)
- Contributed to initial integration of vendor board support into GM's common embedded Linux (Yocto) framework

Silicon Labs, Austin, Texas - *Operations Software - Mixed-Signal/RF Test System Development*

Associate Staff Software Engineer, Feb 2022 - July 2025

- Developed board support software for next-generation semiconductor test systems
- Led SoC (Zynq UltraScale+) evaluation and platform selection
- Ported vendor BSPs and high-speed JESD204B RF ADC FPGA HDL/drivers to a custom Yocto/PetaLinux platform (C, Verilog)
- Developed Linux drivers and APIs for mixed-signal data acquisition and arbitrary waveform generation (C, Python)
- Performed board bring-up, hardware debugging, and root cause analysis using oscilloscopes, logic analyzers, and spectrum analyzers while collaborating with hardware engineers on schematic revisions

NI (formerly National Instruments), Austin, Texas - *Microwave Architectures and Technology Group*

Senior Software Engineer, July 2017 – February 2022

- Developed full Windows driver stack for RF test hardware (C++)
- Developed and debugged calibration algorithms for vector signal transceivers (VSTs) using C++ and LabVIEW
- Diagnosed RF hardware and calibration issues using spectrum analyzers and signal generators
- Partnered with internal engineering teams to resolve complex customer issues across the RF product line
- Selected as 1 of approximately 50 engineers for National Instruments' year-long R&D Self-Leaders leadership development program

BMW Manufacturing Co., Greer, South Carolina - *Software and New Technology Development*

Co-op / Internship, August 2014 – July 2016, Rotational, Every other semester

- Developed control software for automated measurement systems
- Designed C# applications using TCP/IP to control and position measurement devices
- Designed digital communication interfaces and wiring diagrams for servo drive integration
- Enhanced and maintained ASP.NET web applications

Teaching Assistant, University of Florida, Gainesville, FL - *Teaching Assistant*,

Operating Systems Teaching Assistant, Spring 2017, 1 semester

- Assisted students with any issues encountered in the C programming language and Minix operating system

TECHNICAL PROJECTS & ACTIVITIES

- **ESP32 Baby Monitor** – Designed and implemented an embedded baby monitoring and data logging system in MicroPython on ESP32 to track feedings, sleep, and caregiver events; logged over 3,000 real-world events to validate long-term reliability
- **Linux–Zephyr RPMsg IIO Framework** – Developed a custom Linux Industrial I/O (IIO) driver and Zephyr firmware enabling high-throughput data acquisition across heterogeneous ARM SoCs using RPMsg, OpenAMP, shared memory, DMA, and remoteproc
- **FreeRTOS GStreamer Prototype** – Investigated porting GStreamer to FreeRTOS, exploring multimedia pipeline architecture, memory management, and RTOS integration